

MH3100 Real Analysis I

Sample Midterm Test (8 October 2025) (Solutions)

Official sample paper – QRS solutions (Practice 1)

Quantitative Research Society @NTU

October 1, 2026

Overview

- **Coverage:** Lectures 1–17 (real numbers and the Axiom of Completeness, through sequences, series, and sets of real numbers – limit points, closures, closed sets, and (sequential) compactness).
- **Format: FOUR (4)** questions, 6 marks each, 24 marks total. **TIME ALLOWED: 60 minutes. CLOSED BOOK.** Calculators are allowed.
- **Citation policy (verbatim from the paper):** “You may cite any results proved in the lectures and tutorials, unless you are asked by the question to prove the result.”
- **This document:** full QRS solutions to the (SAMPLE) Midterm Test dated 8 October 2025, published as *Practice 1 (Solution by QRS)*.

Instructions to Students (verbatim from the cover page)

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the box provided. Use the **facing page** for more space.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, unless you are asked by the question to prove the result.

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24

Contents

Question 1	(6 Marks)	3
Solution		3
Marking scheme (6 marks)		3
Question 2	(6 Marks)	5
Solution		5
Marking scheme (6 marks)		5
Question 3	(6 Marks)	7
Solution		7
Method 1: both clauses of the definition		7
Method 2: WLOG reduction		7
Marking scheme (6 marks)		7
Question 4	(6 Marks)	9
Solution		9
Method 1: boundedness lemma, factoring		9
Method 2: bound on the tail only		9
Marking scheme (6 marks)		9

Question 1**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks, QRS assumption) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.
- (b) (2 marks, QRS assumption) There is a sequence with a subsequence that is bounded but contains no subsequence that converges.
- (c) (2 marks, QRS assumption) If every proper subsequence of a sequence $\{a_n\}$ converges, then $\{a_n\}$ converges as well.

Solution

- (a) **Proof.** Suppose $\{a_n\}$ and $\{a_n + b_n\}$ converge. $\therefore \{-a_n\}$ converges (ALT, $c = -1$). $\therefore \{(a_n + b_n) + (-a_n)\} = \{b_n\}$ converges (ALT), contradicting that $\{b_n\}$ diverges. $\square$

False

- (b) **Proof.** Let $\{a_{n_k}\}$ be any bounded subsequence of $\{a_n\}$. $\therefore \{a_{n_k}\}$ has a convergent subsequence (BW). $\therefore$ no bounded subsequence of $\{a_n\}$ lacks a convergent subsequence. $\square$

False

- (c) **Proof.** Suppose every proper subsequence of $\{a_n\}$ converges. $b_k := a_{k+1}$ has $n_k = k+1 \neq k \therefore \{b_k\}$ is proper $\therefore L = \lim b_k$ for some $L \in \mathbb{R}$. $c_1 := a_1$, $c_k := a_{k+1}$ ($k \geq 2$) has $n_2 = 3 \neq 2 \therefore \{c_k\}$ is proper $\therefore M = \lim c_k$ for some $M \in \mathbb{R}$. The tail $a_3, a_4, \dots$ is a subsequence of both $\{b_k\}$ and $\{c_k\} \therefore L = M$ (Subsequential Limits, uniqueness). Let $\varepsilon > 0$; choose $K \in \mathbb{N}$ with $|a_{k+1} - L| < \varepsilon$ for all $k \geq K$ (def. of limit). $\therefore |a_n - L| < \varepsilon$ for all $n \geq K + 1$. $\therefore L = \lim a_n$ (def. of limit). $\square$

True

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial): the printed paper awards these 6 marks purely for the three correct True/False answers, since “You do NOT need to justify your answer.” The scheme below is what QRS would use if justification were required, for teaching purposes.

- (a) (2 marks) Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for the answer, 1 mark for correctly invoking the Algebraic Limit Theorem to derive a contradiction.)

- (b) (2 marks) Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for the answer, 1 mark for correctly invoking the Bolzano-Weierstrass Theorem.)

- (c) (2 marks) Correct answer True.

- 2 marks: correct answer. (If justification were required: 1 mark for correctly identifying that deleting a single term produces a proper subsequence, 1 mark for the shared-tail argument forcing $\{a_n\}$ itself to converge.)

Typical deductions if justification were graded: -1 : claims $\{a_n\}$ is itself one of the “proper” subsequences in part (c); -1 : asserts the Bolzano-Weierstrass Theorem without first noting boundedness of the subsequence in part (b); -0.5 : sentence fragments in place of a full justification.

Unofficial – QRS@NTU – qrsntu.org

Question 2**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks, QRS assumption) There are two divergent series $\sum a_n$ and $\sum b_n$ where $\sum a_n b_n$ converges.
- (b) (2 marks, QRS assumption) The set $\{x \in \mathbb{R} : x \neq 0\}$ is closed.
- (c) (2 marks, QRS assumption) If K is a compact set and F is a closed set, then $K \setminus F$ is a compact set.

Solution

- (a) **Proof.** Let $a_n = b_n := 1/n$. $\therefore \sum a_n = \sum b_n = \sum 1/n$ diverges (harmonic series: $\frac{1}{2^{k+1}} + \dots + \frac{1}{2^{k+1}} \geq \frac{1}{2}$ for all $k \in \mathbb{N}$). For $n \geq 2$: $\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$
 $\therefore \sum_{n=2}^N \frac{1}{n^2} \leq 1 - \frac{1}{N} < 1$ for all $N \geq 2$. $\therefore s_N := \sum_{n=1}^N \frac{1}{n^2} \leq 2$ for all $N \in \mathbb{N}$, and $\{s_N\}$ is increasing $\therefore \{s_N\}$ converges (MCT) $\therefore \sum a_n b_n = \sum \frac{1}{n^2}$ converges. $\square$

True

- (b) **Proof.** Let $C := \{x \in \mathbb{R} : x \neq 0\}$ and $a_n := 1/n$. $a_n \in C$ for all $n \in \mathbb{N}$ and $0 = \lim a_n \notin C$. $\therefore C$ is not closed (Sequential Criterion for Closed Sets). $\square$

False

- (c) **Proof.** Let $K := [0, 1]$ and $F := \{0\}$. F is finite $\therefore$ has no limit points $\therefore F$ is closed. K is closed and bounded $\therefore K$ is compact (Sequential Compactness of Closed and Bounded Sets). $K \setminus F = (0, 1] \ni a_n := 1/n$ for all $n \in \mathbb{N}$; every subsequence of $\{a_n\}$ converges to $0 \notin (0, 1]$ (Subsequential Limits). $\therefore (0, 1] = K \setminus F$ is not compact (def. of compact). $\square$

False

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial): as in Question 1, the printed paper awards these 6 marks purely for the three correct True/False answers. The scheme below is what QRS would use if justification were required.

- (a) (2 marks) Correct answer True.

- 2 marks: correct answer. (If justification were required: 1 mark for a valid example with both $\sum a_n, \sum b_n$ divergent, 1 mark for correctly showing $\sum a_n b_n$ convergent, e.g. via the Comparison Test.)

- (b) (2 marks) Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for exhibiting a convergent sequence in C with limit outside C , 1 mark for correctly invoking the Sequential Criterion for Closed Sets.)

- (c) (2 marks) Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for a valid counterexample (K, F) with K compact and F closed, 1 mark for correctly showing $K \setminus F$ fails compactness via an explicit non-convergent-in-the-set sequence.)

Typical deductions if justification were graded: -1 : in part (a), an example where $\sum a_n b_n$ does not actually converge (e.g. forgetting to check the product series, not just the individual terms); -1 : in part (c), a counterexample where F is not actually closed or K is not actually compact; -0.5 : sentence fragments in place of a full justification.

Question 3**(6 Marks)**

In this question, A and B denote nonempty sets of real numbers that are bounded above.

Write down the definition of $s = \sup A$, and use the definition to show that

$$\max\{\sup A, \sup B\} = \sup(A \cup B).$$

Do not use the ε -Criterion for suprema and infima.

(Source: Tutorial 1, Question 2, with the added instructions to first state the definition of supremum and to avoid the ε -Criterion.)

Solution

Definition. $s = \sup A \iff (1) a \leq s$ for all $a \in A$; (2) $s \leq b$ for every upper bound b of A .

Method 1: both clauses of the definition

Proof. $s := \sup A, t := \sup B$ exist (AoC), $M := \max\{s, t\}$. Let $x \in A \cup B$. If $x \in A$: $x \leq s \leq M$ (def. (1)). If $x \in B$: $x \leq t \leq M$ (def. (1)). $\therefore M$ is an upper bound of $A \cup B$. $A \cup B \supseteq A \neq \emptyset$ and is bounded above by $M \therefore \sup(A \cup B)$ exists (AoC). Let b be an upper bound of $A \cup B$. $A, B \subseteq A \cup B \therefore b$ bounds A and B above $\therefore s \leq b$ and $t \leq b$ (def. (2)). $\therefore M = \max\{s, t\} \leq b \therefore M = \sup(A \cup B)$, i.e. $\max\{\sup A, \sup B\} = \sup(A \cup B)$. $\square$

Method 2: WLOG reduction

Proof. $s := \sup A, t := \sup B$ exist (AoC); WLOG $s \geq t$ (symmetry in A, B) $\therefore \max\{s, t\} = s$. $x \in A \implies x \leq s$; $x \in B \implies x \leq t \leq s$ (def. (1)) $\therefore s$ is an upper bound of $A \cup B \supseteq A \neq \emptyset$. $\therefore \sup(A \cup B)$ exists (AoC). An upper bound b of $A \cup B$ bounds A above $\therefore s \leq b$ (def. (2)) $\therefore s = \sup(A \cup B)$. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial).

- (i) **(1 mark) Definition.** The definition of $s = \sup A$ stated correctly, with both the “upper bound” and “least upper bound” clauses, and quantifiers in the correct order (“for all $a \in A, a \leq s$ ”; “for every upper bound b of $A, s \leq b$ ”).
 - -0.5 : only one of the two clauses given.
 - -0.5 : quantifier order reversed or omitted (e.g. writing “ $s \leq b$ for some upper bound b ”).
- (ii) **(1 mark) Existence.** Noting that $A \cup B$ is nonempty and bounded above, so that $\sup(A \cup B)$ exists by the Axiom of Completeness, before using it.
 - -1 : uses $\sup(A \cup B)$ without ever justifying its existence.
- (iii) **(2 marks) Upper bound direction.** A correct proof that $\max\{\sup A, \sup B\}$ is an upper bound for $A \cup B$, with the case split $x \in A / x \in B$ (or the WLOG reduction of Method 2) fully carried out.
 - -1 : case split incomplete (only one of A, B checked).

(iv) **(1.5 marks) Least-upper-bound direction.** A correct proof that $\max\{\sup A, \sup B\}$ is not greater than any upper bound of $A \cup B$, via an arbitrary upper bound b of $A \cup B$, restricting b to A and to B , and applying the definition of $\sup A$ and $\sup B$ to conclude $\sup A \leq b$ and $\sup B \leq b$.

- -1 : asserts $\sup A \leq \sup(A \cup B)$ (or similar) without the arbitrary-upper-bound argument.

(v) **(0.5 marks) Writing.** Full sentences, connectives, and inference words (“therefore”, “hence”, “so”) throughout, rather than a bare symbolic chain.

Typical further deductions: -1 : uses the ε -Criterion for suprema and infima, which is explicitly banned by the question; -0.5 : sentence fragments or unjustified jumps between lines.

Question 4

(6 Marks)

Write down the definition of a Cauchy sequence $\{a_n\}$, and **use the definition to show** that if $\{a_n\}$ is a Cauchy sequences [sic], then $\{a_n^2\}$ is a Cauchy sequence.

Do not use the Cauchy Criterion or the Algebraic Limit Theorem.

(Source: Tutorial 2, Question 5(b), with the added instruction to first state the definition of a Cauchy sequence.)

Solution

Definition. $\{a_n\}$ is Cauchy $\iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n, m \geq N : |a_n - a_m| < \varepsilon$.

Method 1: boundedness lemma, factoring

Lemma. $\{a_n\}$ Cauchy $\implies \exists M > 0$ with $|a_n| \leq M$ for all $n \in \mathbb{N}$.

Proof of Lemma. Take $\varepsilon = 1$: choose N with $|a_n - a_m| < 1$ for all $n, m \geq N$. For all $n \geq N$:

$$|a_n| \leq |a_n - a_N| + |a_N| < 1 + |a_N| \quad (\text{triangle inequality})$$

$M := \max\{|a_1|, \dots, |a_{N-1}|, 1 + |a_N|\} \geq 1 > 0$ (finite list) $\therefore |a_n| \leq M$ for all $n \in \mathbb{N}$. $\square$

Proof of Q4. Let $M > 0$ with $|a_n| \leq M$ for all $n \in \mathbb{N}$ (Lemma). Let $\varepsilon > 0$; choose N with $|a_n - a_m| < \frac{\varepsilon}{2M}$ for all $n, m \geq N$ (Cauchy). For all $n, m \geq N$:

$$\begin{aligned} |a_n^2 - a_m^2| &= |a_n - a_m| |a_n + a_m| & (a_n^2 - a_m^2 &= (a_n - a_m)(a_n + a_m)) \\ &\leq |a_n - a_m| (|a_n| + |a_m|) \leq 2M |a_n - a_m| & (\text{triangle inequality; } |a_n|, |a_m| \leq M) \\ &< 2M \cdot \frac{\varepsilon}{2M} = \varepsilon \end{aligned}$$

$\therefore \{a_n^2\}$ is Cauchy (def. of Cauchy). $\square$

Method 2: bound on the tail only

Proof. Let $\varepsilon > 0$; choose N_0 with $|a_n - a_m| < 1$ for all $n, m \geq N_0$ (Cauchy, with 1 in place of ε). $\therefore |a_n| \leq |a_n - a_{N_0}| + |a_{N_0}| < 1 + |a_{N_0}| =: M$ for all $n \geq N_0$ (triangle inequality). Choose $N \geq N_0$ with $|a_n - a_m| < \frac{\varepsilon}{2M}$ for all $n, m \geq N$ (Cauchy). For all $n, m \geq N$:

$$\begin{aligned} |a_n^2 - a_m^2| &= |a_n - a_m| |a_n + a_m| \leq |a_n - a_m| (|a_n| + |a_m|) & (\text{triangle inequality}) \\ &< \frac{\varepsilon}{2M} \cdot 2M = \varepsilon & (|a_n|, |a_m| < M) \end{aligned}$$

$\therefore \{a_n^2\}$ is Cauchy (def. of Cauchy). $\square$

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial).

- (i) **(1 mark) Definition.** The definition of a Cauchy sequence stated correctly, with quantifiers in the correct order: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < \varepsilon$.

- -0.5: quantifier order wrong (e.g. N chosen before ε , or N allowed to depend on n).

- -0.5 : uses the single-index form ($|a_n - a_N| < \varepsilon$) without adapting the rest of the proof to match, or silently swaps between the two forms.
- (ii) **(1.5 marks) Boundedness Lemma.** A correct, self-contained proof that a Cauchy sequence is bounded, directly from the Cauchy definition with $\varepsilon = 1$, splitting into the tail (bounded by $1 + |a_N|$) and the finitely many head terms.
- -1 : boundedness asserted without proof, or proved via convergence/the Cauchy Criterion.
 - -0.5 : omits the finitely-many-head-terms case, or does not verify $M > 0$.
- (iii) **(2 marks) Factoring estimate.** The identity $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$, the triangle-inequality bound $|a_n + a_m| \leq |a_n| + |a_m| \leq 2M$, and the resulting inequality $|a_n^2 - a_m^2| \leq 2M |a_n - a_m|$, all correctly derived.
- -1 : factoring identity missing or incorrect.
 - -0.5 : bound $|a_n + a_m| \leq 2M$ not justified from the boundedness of $\{a_n\}$.
- (iv) **(1 mark) Assembly.** The correct choice $\varepsilon' := \varepsilon/(2M)$ (or equivalent), applied to the Cauchy hypothesis, and the final chain of inequalities correctly matched back to the definition of $\{a_n^2\}$ being Cauchy.
- -0.5 : N chosen depending on n (or m), rather than depending only on ε .
- (v) **(0.5 marks) Writing.** Full sentences, connectives, and inference words throughout; quantifiers written in words (“let $\varepsilon > 0$ be given,” “there exists $N \in \mathbb{N}$ such that”), never as a bare symbolic string.

Typical further deductions: -1 : uses the Cauchy Criterion or the Algebraic Limit Theorem, both explicitly banned by the question; -0.5 : sentence fragments / unjustified jumps between lines.